Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 2 – Project Management Plan

– FPT University, October 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
27/10/2026	M	Lê Nguyễn Quang Đức	Remove template residue (MySQL/GitLab/Eclipse), fill actual technology stack versions and deliverables, and align diagram inventory to 23 pages.

*A - Added M - Modified D - Deleted

II. Project Management Plan

1. Overview

1.1 Scope & Estimation

Workstreams: WF-01 school/robot allocation; WF-02 classroom learning and submission; WF-03 schematic experiment and runtime integration; WF-04 assessment/AI metadata. Effort estimates remain team estimates pending sign-off, not timesheets.

Effort and complexity are estimated by workflow and technical risk; the official Project Tracking WBS is maintained in its existing WBS, Issues, Defects and Q&A tabs.

#	WBS Item	Complexity	Est. Effort (man-days)
1	Access/organization	Medium (planning estimate)	Estimate pending team validation
2	Learning and assignment	Medium (planning estimate)	Estimate pending team validation
3	Experiment/runtime	Medium (planning estimate)	Estimate pending team validation
4	Assessment/AI metadata	Medium (planning estimate)	Estimate pending team validation
5	SRS and SDD	Medium (planning estimate)	Estimate pending team validation
6	CI/test evidence	Medium (planning estimate)	Estimate pending team validation
		Medium (planning estimate)	Estimate pending team validation

1.2 Project Objectives

Review 2: trace Registration v1.2 to SRS/SDD, code/schema and test plan. Defense 1.1: demonstrate four main workflows with main and exception paths on declared staging, using documented targets rather than fabricated passes.

Objectives: demonstrate a reproducible Iteration 1, maintain consistency among Registration, SRS, SDD, migrations, tests and CI. Targets for Defense 1.1 are recorded in BM01 Appendix A and become binding only after Review 2.

Quality

#	Testing Stage	Test Coverage	No. of Defects	% of Defect	Notes
1	Reviewing	SRS, SDD, Architecture & ERD diagrams	0	0%	
2	Unit Test	Education API (NestJS) & Web (Vitest)	0	0%	
3	Integration Test	Database constraints & scope authorization	0	0%	
4	System Test	4 Main Workflow demonstrations (Defense 1.1 target)	0	0%	
5	Acceptance Test	Mandatory scenario checklist (Defense 1.1 target)	0	0%	

Quality: authorization denial must be tested; physical safety must fail closed; no invented experiment execution success. Track actual test counts separately.

Milestone timeliness: to be measured against supervisor-approved schedule; no achieved percentage asserted.

Allocated effort: owner estimates to be approved; no fabricated actual effort.

1.3 Project Risks

Major risks: mismatched ERD cardinality/state transitions, physical robot integration gaps, insufficient CI/test evidence and multi-organization authorization leakage.

#	Risk Description	Impact	Possibility	Response Plans
1	Physical runtime integration unverified	High	High	Separate schematic request from hardware execution and collect tested E2E evidence
2	Cross-school authorization and multi-tenant data leakage	High	Medium	Enforce strict JWT validation, organization/classroom scope guards, and automated negative test suites
3	Diagram states and ERD constraints mismatch implementation	High	Medium	Perform rigorous cross-document audit against Prisma schema, migrations, and NestJS service enums

2. Management Approach

Iterative development across four owner-led workflows. Design revisions and code changes use PR review; Registration scope is frozen only after Review 2 approval.

2.1 Project Process

Iterative delivery, feature branches/PR review, synchronized FR-to-code-to-test changes, review-2 Git tag for immutable self-review baseline.

Process: Requirements → SRS/SDD and diagram review → feature branch implementation → unit/integration tests → PR/CI → staging demo → issue tracking.

2.2 Quality Management

Review each feature against registration, actor, scope guard, state path, database constraints and test cases. Never mark a test Passed without observed run evidence.

- Prevent defects with validation, least-privilege scope tests, immutable program versions and verified migrations.
- Review PR changes and diagram/state alignment prior to merge.
- Unit tests: NestJS Jest, frontend Vitest, Python runtime tests where available.
- Integration tests: API/DB authorization and attempt–program–submission invariants.

- System tests: the four workflow main and exception demonstrations using seeded staging data.

2.3 Training Plan

Learning topics: hardware safety, Prisma migrations, Supabase membership authorization, CI workflows and test traceability; plan training as gaps are confirmed.

Training Area	Participants	When, Duration	Waiver Criteria
Hardware safety & tabletop robot control protocols	All members	Week 4-5, 8 hours	Mandatory
Prisma schema migrations & PostgreSQL transaction locking	Đức, Lợi	Week 3, 6 hours	Mandatory
Git feature branch workflow & CI test automation	All members	Week 2, 4 hours	Mandatory

3. Project Deliverables

Registration v1.2 + BM03, reports 1–5, BM01/BM02, 4 workflow sources and C4/ERD/state diagrams, seed/database script, CI/test evidence and project tracking. BM04 due after Review 2.

Deliverables: registration with BM03, Reports 1–5, Project Tracking, PlantUML/Mermaid diagrams, SQL migrations, code and test evidence, BM02 and BM01 Appendix A.

#	Deliverable	Due Date	Notes
1	Registration Form v1.2 & Record of Changes BM03	Review 2	Formal project scope baseline
2	Reports 1-5 (Introduction, Plan, SRS, SDD, Test Documentation)	Review 2	Complete technical documentation package
3	Design Brief BM01 & Self-Review Form BM02	Review 2	Official university evaluation forms
4	23-Page	Review 2	C4 Level 1-3, 4 Workflows, 4 State

	Authoritative Architecture & ERD Diagrams		Machines, 12 ERDs
5	PostgreSQL Database Migrations & Seed Script	Review 2	Reproducible schema and test data
6	Automated Test Suites & Execution Evidence Record	Review 2	Jest & Vitest test suites with local verification log
2	Reports 1-5 (Introduction, Plan, SRS, SDD, Test Documentation)	Review 2	Complete technical documentation package
3	Design Brief BM01 & Self-Review Form BM02	Review 2	Official university evaluation forms
4	23-Page Authoritative Architecture & ERD Diagrams	Review 2	C4 Level 1-3, 4 Workflows, 4 State Machines, 12 ERDs
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4. Responsibility Assignments

Decisions and workflow ownership: Đức WF-01/auth/organization; Đạt WF-02/curriculum/submission; Tuấn WF-03/robot runtime; Lợi WF-04/AI metadata/assessment. Real delivery and PR assignments must be verified.

Ownership: Đức WF-01 and integration; Đạt WF-02 and course delivery; Tuấn WF-03 and robot runtime; Lợi WF-04, assessment and ERD.

D~Do; R~Review; S~Support; I~Informed; <blank>- Omitted

Responsibility	Lê Nguyễn Quang	Trần Tiến Đạt	Vũ Anh Tuấn	Phạm Viết Lợi	ThuyVT04278
WF-01	D	R	R	R	R
WF-02	R	D	R	R	I
WF-03	R	R	D	R	R
WF-04	R	R	R	D	R
SRS/SDD	R	R	R	D	

5. Project Communications

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Weekly Team Standup	All members	Review progress, resolve blockers, assign PRs	Weekly, 1 hour	In-person / Discord
Advisor Sync Meeting	Team & Supervisor	Report progress, receive feedback, approve scope	Bi-weekly, 1 hour	FPT University / Google Meet
Code & Document PR Review	Team members	Peer review changes before merging into main	Continuous on PR	GitHub PRs

Weekly meetings; all members attend Review 2, prepare code explanations and ownership evidence.

6. Configuration Management

6.1 Document Management

Keep Word/Excel on official university templates, diagrams sourced from user-approved PDF, editable source diagram versions in repo; use SEP490_FA26SE183_R2_ filenames and seven-folder submission structure.

Version Word/PDF documents with project code FA26SE183; editable diagrams must correspond to repository source. Maintain explicit Draft versus Approved status.

6.2 Source Code Management

Source repository https://github.com/SKYNET-Thesis/se (main); do not tag review-2 until CI, docs and schema have been checked. BM02 must analyze exact tagged commit.

Repository: https://github.com/SKYNET-Thesis/se. Work in feature branches, review via PR; tag review-2 only after code/documents are reconciled. Never rewrite tag after official self-review.

Technology: React 18/Vite, NestJS 11, Prisma/PostgreSQL, Supabase Auth, Python Robot Runtime, Docker, GitHub Actions. AWS staging is planned and must not be reported as deployed without evidence.

Category	Tools / Infrastructure
Technology	React 18, Vite 5, NestJS 11, TypeScript 5.9, Python 3.10+
Database	PostgreSQL 15+, Supabase, Prisma ORM 5.21
IDEs/Editors	Visual Studio Code, Cursor
Diagramming	Diagrams.net (Draw.io), Mermaid
Documentation	Microsoft Word, Microsoft Excel, Markdown
Version Control	Git, GitHub (Repository: SKYNET-Thesis/se)
Testing Tools	Jest 30.5, Vitest 2.1, Playwright
Deployment / Staging	Docker, GitHub Actions (AWS ECS/RDS planned for Defense 1.1)

Review 2 design verification milestone: validate 23-page diagram source (3 C4, 4 Activity, 4 State Machines, 1 Conceptual ERD, 5 Logical ERDs (including separate 16a and 16b), 6 Physical ERDs), align Report 3/4/5 references, then review DB constraints and evidence prior to review-2 tag.